# QuickU

# Demand Forecasting & Inventory Planning

## From Demand Patterns to Actionable Inventory Priorities

**DATA-DRIVEN**

Forecasting with Historical Insights

**ACTIONABLE**

Inventory Planning Across Hub & Categories

**ACCURATE**

Methodology & Validation Based

**BUSINESS**

Impact on Product Availability & Customer Satisfaction

Forecast for
**August 2022**

Analysis Level
**Hub Level x Week**

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# Business Problem & Objectives

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Turning demand patterns into better inventory planning decisions

## THE CHALLENGE

QuickU needs to anticipate demand for August 2022 to prepare inventory across hubs and categories.

Historical demand shows different patterns across weeks, hubs, and categories, making a reliable demand forecast essential for inventory planning.

DEMAND UNCERTAINTY

↓

FORECAST

↓

INVENTORY PLANNING

## WHAT WE AIM TO ANSWER

**01 Select the right forecasting method**
Compare forecasting approaches based on validation performance.

**02 Forecast August 2022 demand**
Estimate weekly demand to support inventory preparation.

**03 Identify inventory priorities**
Translate the forecast into actionable hub & category priorities.

## DATA SCOPE

**Jan - Jul 2022**
Historical period

**5 Hubs**
A • B • C • D • E

**10 Categories**
A - J

**Weekly Demand**
Shipped Quantity

# Analytical Approach

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## From data to actionable inventory planning

**01 Data Collection**
Jan–Jul 2022 historical weekly demand data by Hub & Category

**02 Exploratory Data Analysis (EDA)**
Trend, weekly seasonality, demand by hub & category

**03 Model Comparison**
Compare 3 forecasting models using MAE, RMSE, and MAPE

**04 Demand Forecasting**
Forecast weekly demand for August 2022 using the best model

**05 Inventory Planning**
Translate forecast into hub & category priorities for stock preparation

### WHAT WE ANALYZE

**Demand Pattern**
Overall trend & weekly seasonality

**Category Performance**
Identify top & low demand categories

**Hub Performance**
Compare demand across hub

**Inventory Risk**
Assets category volatility to prioritize inventory

### WHAT WE ANALYZE

**Python**
Data Processing & Modeling

**Pandas & NumPy**
Data Manipulation & Analysis

**Moving Average (4-Week)**
Selected Forecasting Model

**Power BI**
Dashboard & Visualization

**Expected Outcome:** ✓ Accurate forecast of August 2022 weekly demand ✓ Clear visibility of demand by hub and category ✓ Actionable inventory priorities to reduce stockout risk and overstock

# Forecasting Methodology

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## Comparing forecasting models to identify the best approach for August 2022

### Models Compared

| | Model 1<br>4-Week Moving Average | Model 2<br>Linear Regression | Model 3<br>Seasonal Linear Regression |
|---|---|---|---|
| **Description** | Forecasts demand based on the average demand of the previous 4 weeks. | Models the overall linear trend in historical demand. | Combines linear trend with weekly seasonality. |
| **How it Works** | Calculates the average of the previous 4 weeks and uses it as the next forecast. | Fits a straight line to historical demand over time. | Fits a trend while incorporating recurring weekly patterns. |
| **Strength** | Simple and responsive to recent demand changes. | Captures long-term trend. | Captures both trend and recurring weekly patterns. |
| **Limitation** | May not fully capture strong seasonality or long-term trend changes. | Less effective when demand contains seasonal patterns. | More complex and requires sufficient seasonal pattern stability. |

We compared 3 forecasting models and selected the best model based on validation performance on July 2022 data.

### Validation Results (July 2022)

| Model | MAE (units) | RMSE (units) | MAPE (%) |
|---|---|---|---|
| **4-Week Moving Average** | **29,030.75** | **36,337.99** | **31.05** |
| Seasonal Linear Regression | 30,082.27 | 31,539.05 | 38.13 |
| Linear Regression | 44,816.88 | 49,156.62 | 59.31 |

### Best Model Selection

**4-Week Moving Average**
Selected as the **best model** based on the lowest MAE, RMSE, and MAPE.

- Lowest MAE
- Lowest RMSE
- Lowest MAPE
- Simple & easy to implement

Therefore, the 4-Week Moving Average was selected to forecast August 2022 weekly demand.

# Historical Demand Trend

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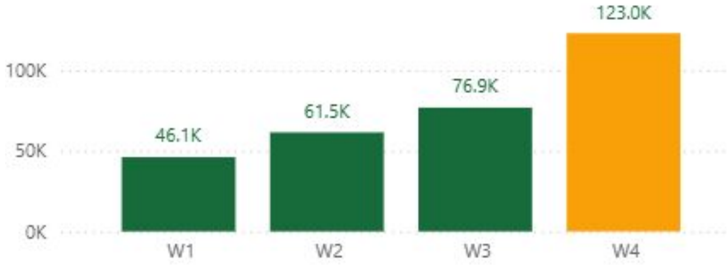
## January – July 2022 Weekly Demand Overview

### Monthly Demand Trend (Jan - Jul 2022)


Overall demand shows an increasing trend.

### Historical Weekly Demand Pattern (Jan - Jul 2022)




W4 is the peak demand period.

**1. Increasing Trend**

Overall demand increased throughout the observation period.

**2. Weekly Seasonality**

Demand consistently rises from W1 to W4, indicating a clear weekly pattern.

**3. Forecasting Implication**

Both trend and weekly patterns should be considered when forecasting August demand.

**Key Takeaway**: Demand shows both an increasing trend and recurring weekly seasonality, providing the basis for comparing forecasting models.

# August 2022 Forecast

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## August demand is forecasted to increase progressively, peaking in W4

The selected 4-Week Moving Average model is used to forecast August 2022 demand based on historical demand patterns from January–July 2022.

### Total August Forecast

**442.91 K**

units

### Weekly Forecast for August 2022


### Key Insights

- Demand is forecasted to **increase progressively** throughout August.
- W4 is expected to reach **120.1K** units, the highest forecast among all weeks.
- Inventory planning should prepare for higher demand toward **W4**.

### Summary

August 2022 demand is forecasted at approximately 442.9K units, with demand increasing consistently from W1 to W4 and peaking in W4.

The selected 4-Week Moving Average model is used to forecast August 2022 demand based on historical demand patterns from January–July 2022.

# Demand Drivers: Category & Hub

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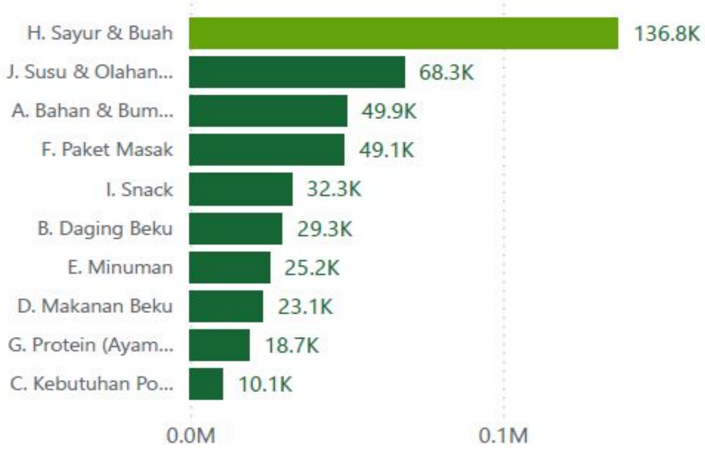
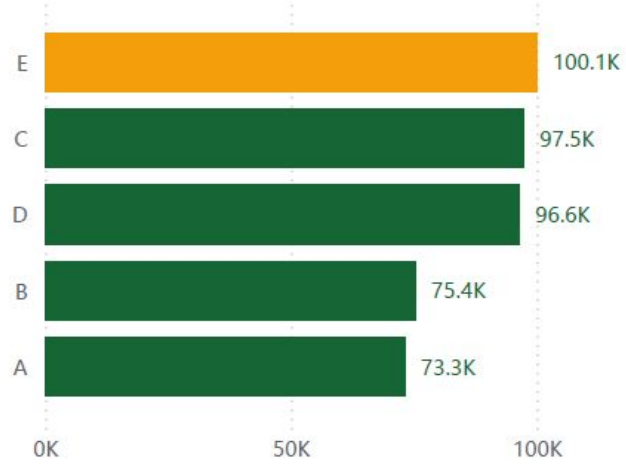
## Identifying where August demand is most concentrated

**Top Categories – August Forecast**




H. Sayur & Buah is the main demand driver, contributing 30.89% of historical average demand.

**Top Hubs – August Forecast**




Hub E, C, and D contribute the largest share of demand.

**Key Takeaway**

Demand is concentrated in H. Sayur & Buah and across Hub E, C, and D, making these areas the primary focus for inventory planning.

# Inventory Risk & Priority

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## High demand does not always mean high inventory risk

| Category | Average Share (%) | August W1 | August W2 | August W3 | August W4 | CV (%) | Priority |
|---|---|---|---|---|---|---|---|
| J. Susu & Olahan Susu | 15.40 | 15,179.56 | 16,697.56 | 17,835.98 | 18,500.07 | 21.06 | 🔴 Critical |
| H. Sayur & Buah | 30.89 | 30,447.14 | 33,491.95 | 35,775.39 | 37,107.41 | 17.17 | 🟠 High |
| F. Paket Masak | 11.09 | 10,930.62 | 12,023.72 | 12,843.48 | 13,321.68 | 23.28 | 🟠 High |
| D. Makanan Beku | 5.22 | 5,144.66 | 5,659.15 | 6,044.98 | 6,270.05 | 27.59 | 🟠 High |
| C. Kebutuhan Pokok | 2.30 | 2,266.42 | 2,493.07 | 2,663.04 | 2,762.19 | 84.15 | 🟠 High |
| A. Bahan & Bumbu Masak | 11.27 | 11,104.97 | 12,215.50 | 13,048.34 | 13,534.17 | 9.91 | 🟡 Medium |
| I. Snack | 7.28 | 7,180.01 | 7,898.04 | 8,436.51 | 8,750.63 | 19.75 | 🟡 Medium |
| B. Daging Beku | 6.62 | 6,528.01 | 7,180.83 | 7,670.41 | 7,956.00 | 12.62 | 🟡 Medium |
| E. Minuman | 5.70 | 5,615.77 | 6,177.36 | 6,598.53 | 6,844.21 | 18.94 | 🟡 Medium |
| G. Protein (Ayam & Telur) | 4.22 | 4,163.35 | 4,579.70 | 4,891.93 | 5,074.07 | 18.19 | 🟢 Low |

**LOW DEMAND**
**≠**
**LOW INVENTORY RISK**

**C. Kebutuhan Pokok**

Only 2.30% demand share, but **84.15%** CV.

Despite its relatively small demand contribution, its highly volatile demand requires **closer monitoring**.

**Key Takeaway** | Inventory priority should consider both demand contribution and demand volatility – not demand volume alone.

# Hub Analyst

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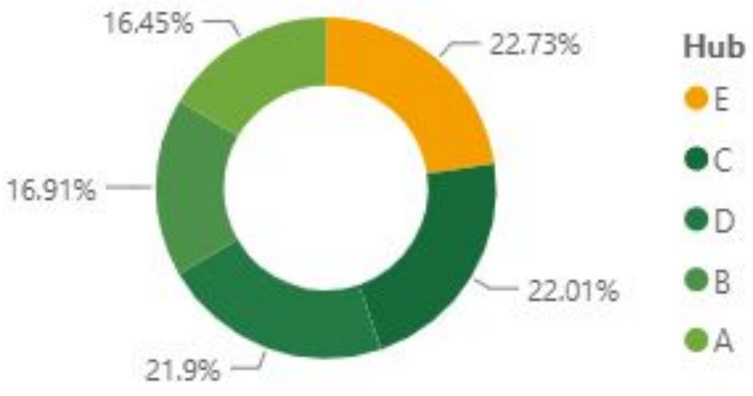
## Hub demand is relatively stable, with the largest contribution coming from Hub E, C, and D.

### Hub Demand Share




E, C, and D contribute **66.64%** of total demand.

### Hub Stability (CV)

| Hub | CV (%) | Interpretation |
|---|---|---|
| E | 1.77% | Very Stable |
| C | 1.28% | Very Stable |
| D | 1.38% | Very Stable |
| B | 2.41% | Very Stable |
| A | 2.47% | Very Stable |

All hubs have **CV below 3%** indicating stable demand distribution.

**Key Takeaway**

Hub E, C, and D contribute the largest demand, while all hubs show very stable demand distribution with CV below 3%.

# Hub x Category & Allocation Validation

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## Translating overall demand into actionable hub x category allocation

| Category | A | B | C | D | E |
|---|---|---|---|---|---|
| A. Bahan & Bumbu Masak | 8,186.19 | 11,465.98 | 14,324.61 | 7,770.31 | 8,177.51 |
| B. Daging Beku | 8,629.38 | 9,066.35 | 4,317.17 | 3,450.57 | 3,882.88 |
| C. Kebutuhan Pokok | 2,871.03 | 2,872.75 | 1,591.48 | 1,467.81 | 1,337.15 |
| D. Makanan Beku | 6,486.79 | 6,860.13 | 2,897.65 | 3,613.72 | 3,252.70 |
| E. Minuman | 5,049.52 | 4,705.21 | 5,041.36 | 5,389.64 | 5,057.25 |
| F. Paket Masak | 8,002.38 | 11,978.97 | 12,753.54 | 7,593.61 | 8,795.81 |
| G. Protein (Ayam & Telur) | 3,737.69 | 3,111.92 | 3,740.64 | 4,372.73 | 3,750.97 |
| H. Sayur & Buah | 17,227.23 | 13,788.81 | 28,627.72 | 37,146.19 | 39,980.53 |
| I. Snack | 3,808.04 | 3,802.05 | 8,523.54 | 8,537.62 | 7,591.53 |
| J. Susu & Olahan Susu | 9,340.21 | 7,787.87 | 15,641.88 | 17,221.63 | 18,278.19 |

| Week | Overall | Hub x Category | Match |
|---|---|---|---|
| W1 | 98,560.50 | 98,560.50 | ✓ |
| W2 | 108,416.88 | 108,416.88 | ✓ |
| W3 | 115,808.59 | 115,808.59 | ✓ |
| W4 | 120,120.49 | 120,120.49 | ✓ |
| **Total** | **442,906.46** | **442,906.46** | ✓ |

**Key Takeaway**

The August forecast of 442.9K units is fully allocated from Overall → Hub → Category → Week, enabling more granular inventory planning.

# Demand Planning Dashboard

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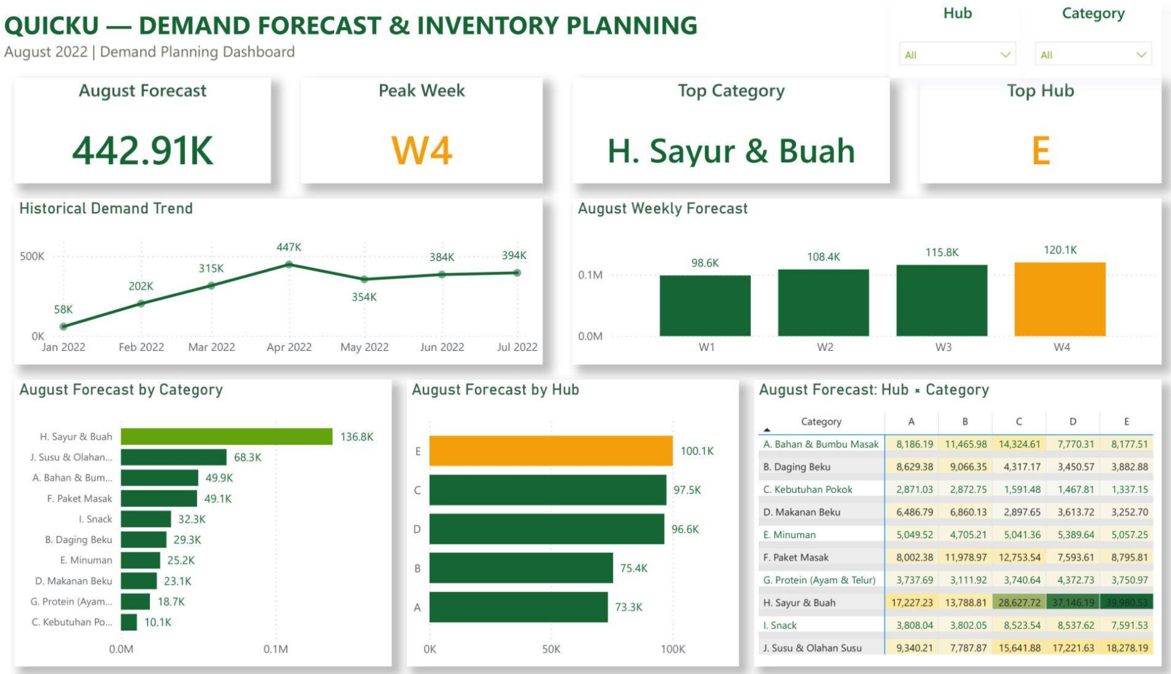
## Power BI dashboard for monitoring August 2022 demand forecast and inventory priorities




| Category | A | B | C | D | E |
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| A. Bahan & Bumbu Masak | 8,186.19 | 11,465.98 | 14,324.61 | 7,770.31 | 8,177.51 |
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**Key Takeaway** | The dashboard translates the forecast into weekly, category, and hub-level insights to support more granular and effective inventory planning.

# External Factor Context

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## Possible external factors that may influence demand movement

**Our analysis is based on online transaction data only.**
The following external factors are potential contributors to demand fluctuation but cannot be validated due to the absence of offline transaction data.

### Ramadhan & Eid Al-Fitr Impact (Apr - May 2022)

- Ramadan (April–May 2022) and Eid al-Fitr may influence grocery demand.
- The demand spike observed in April 2022 aligns with this seasonal period.
- Consumers may stock up for meal preparation and celebrations.

### PPKM Impact (Jul - Aug 2022)

- In July 2022, Indonesia implemented PPKM Level 1–3.
- Restricted mobility may have increased reliance on online grocery shopping.
- This may have contributed to the demand spike observed in July and early August.

**These factors provide context for historical demand patterns and support our forecasting assumptions for August 2022.**

**Key Takeaway**

External factors such as PPKM restrictions and Ramadan–Eid seasonality are potential contributors to demand fluctuations, helping us interpret historical trends and strengthen our forecasting assumptions for August 2022.

External Factors → Demand Impact → Better Forecast Assumptions

# Final Recommendations

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## Actionable steps to optimize inventory planning and fulfillment

| Recommendation | Business Impact |
|---|---|
| **1 Prioritize High-Demand Categories**<br>• Focus inventory and promotions on top-performing categories: H. Sayur & Buah, F. Paket Masak, J. Susu & Olahan Susu, A. Bahan & Bumbu Masak. | **Maximize sales opportunity by** ensuring item availability for high-demand products. |
| **2 Strengthen Inventory in Hub E & D**<br>• Allocate more stock to Hub E and D, where demand is highest.<br>• Monitor weekly forecast to adjust stock dynamically. | **Reduce stockout risk** in high-demand hubs and improve customer satisfaction. |
| **3 Plan Ahead for External Factors**<br>• Anticipate demand changes during PPKM periods and future Ramadan & Eid seasons.<br>• Adjust inventory planning proactively. | **Improve forecast accuracy** and inventory preparedness during high-impact periods. |
| **4 Reduce Expired Stock Risk**<br>• Implement inventory rotation based on demand forecast.<br>• Use demand-based allocation across hubs and categories. | **Minimize waste** and improve inventory efficiency and profitability. |
| **5 Data-Driven Continuous Monitoring**<br>• Monitor demand trend, category performance, and hub metrics regularly.<br>• Use dashboard insights to support faster decision-making. | **Enable quick decision making** and adaptive inventory planning based on real-time insights. |

**Key Takeaway** | By executing these recommendations, QuickU can optimize inventory allocation, reduce waste and stockouts, and meet customer demand more effectively.

# Limitations

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## Current analysis constraints and data gaps

**Our analysis is based on online transaction data only.**
The following data are not available, which limits our ability to determine the exact purchase quantity for inventory planning.

### NO CURRENT STOCK DATA
We do not have real-time stock information across hubs and categories.

**Why it matters:**
Without current stock, we cannot measure inventory position or determine how much stock should be replenished.

### NO SAFETY STOCK INFORMATION
Safety stock level for each category and hub is not available.

**Why it matters:**
Safety stock is essential to buffer demand uncertainty and prevent stockouts.

### NO LEAD TIME INFORMATION
We do not have supplier lead time for restocking inventory.

**Why it matters:**
We do not have supplier lead time for restocking inventory.

### NO REORDER POINT CALCULATION
Without stock, safety stock, and lead time, we cannot calculate the exact reorder point.

**Why it matters:**
Reorder point is the key trigger for purchasing decisions to avoid stockouts or overstock.

### NO OFFLINE SALES DATA
Our dataset only includes online transactions; offline sales are not included.

**Why it matters:**
Total demand may be underestimated if offline demand is significant.

### IMPACT TO PURCHASE QUANTITY
Due to these limitations, we are unable to recommend the exact purchase quantity. Our forecast indicates demand and helps identify priorities, but further data is needed for precise inventory decision-making.

### What we can do:
✓ Forecast demand and identify high-demand categories and hubs
✓ Prioritize inventory allocation based on demand insights
✓ Monitor performance and adjust planning continuously

**Key Takeaway** | Additional data on stock, safety stock, lead time, reorder point, and offline sales will enable more accurate purchase quantity planning and better inventory decisions.

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# THANK YOU!!

Dashboard
Power BI

Notebook
Google Collab

GitHub